

WHY

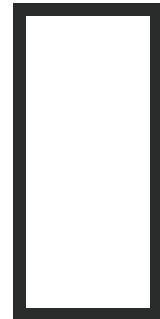
The canonical statement of **why** conception-space exists — the intent any feature, design decision, or refactor should serve. Not a roadmap, not a feature list, not a how-to. **northstar.md** is the source of truth; this is its hard-edge presentation.

CONCEPTION-SPACE · INTENT

Northstar

**Organize your knowledge in space.
Build navigable places. Shape visible
relationships. Discover emergent
patterns.**

It is the **intent** — the underlying purpose that any specific feature, design decision, or refactor should be in service of.

**USE IT LIKE THIS**

- When proposing a new feature, check whether it serves the intent below. If it doesn't, it probably doesn't belong.
- When choosing between two implementations, prefer the one that better preserves the intent.

- When working becomes confused about "what is this even for," return here.
- When the work surfaces a new design principle that follows from the intent, add it as an axiom below so future work inherits it.

01

THE INTENTS

Four complementary facets of the project's purpose. The *why*, in broad strokes.



INTENT 1

Organize your knowledge in space

Gives knowledge spatial structure the mind can remember.



INTENT 2

Build navigable places

Spatial navigation helps knowledge become easier to revisit.



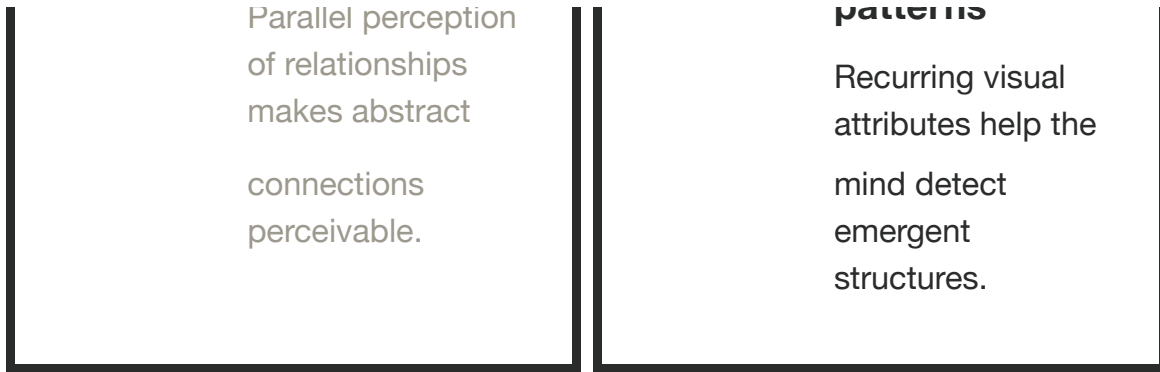
INTENT 3

Shape visible relationships



INTENT 4

Discover emergent patterns



1

Organize your knowledge in space

Human memory and perception are strongly spatial. A place gives knowledge landmarks, regions, paths, distances, and orientations, which help the mind recognize relationships instead of having to recall everything abstractly. We remember locations, regions, landmarks, and relationships well, and we use spatial structure to reduce mental load. When knowledge is organized in space, we can recognize patterns, revisit familiar areas, compare nearby ideas, notice gaps, and build a cognitive map of what we know.

2

Build navigable places

Navigable places have cognitive value because humans are especially good at remembering and reasoning with spatial structure. Navigation turns understanding into an active process: we can move through ideas, return to familiar areas, compare nearby concepts, and gradually form a cognitive map. In this sense, the space becomes an external memory aid that reduces cognitive load and makes complex relationships easier to perceive, revisit, and develop.

3

Shape visible relationships

Visible relationships are valuable because cognition depends heavily on perceiving relations, not just isolated things. When relationships are made visible through distance, grouping, alignment, connection, containment, motion, color, or scale, we can recognize patterns directly instead of reconstructing them from text or memory. Shaping those relationships also turns understanding into an active process: we can adjust a representation, test whether it "fits", notice inconsistencies, and refine the structure. This supports perception, memory, attention, and reasoning by making abstract relations external, manipulable, and easier to compare.

4

Discover emergent patterns

When many ideas and relationships are arranged visibly, emergent structures are revealed that were not obvious item by item: clusters, gaps, bridges, symmetries, outliers, repeated forms and motions, and unexpected analogies.

This matters because our perception is good at detecting patterns across space. By externalizing knowledge into a visible arrangement, our perceptual system participates in thinking: the eye can notice structure before the mind has to verbalize it. That supports memory, attention, and reasoning because we can recognize higher-order organization, see what is missing or overconnected, and form a stronger cognitive map of the knowledge.

02

CONCEPTS

The intents say *why* in broad strokes. These concepts are their underlying mechanisms and the source of the axioms below.

1

Parallel perception of relationships

conception-space is for parallel perception of relationships.

A list, an index, or an appendix forces relationships through a *linear* perceptual channel — one reads item by item, holding the previous in working memory while one reads the next, building the structure incrementally and forgetting most of it by the end. Even with a great index, "what is the *shape* of these 200 relationships?" is unanswerable from the index itself; one has to construct that shape in one's head, one relationship at a time.

A visual space pushes the same information through a *parallel* perceptual channel — one's visual system, which is built exactly for this. The moment one looks at the scene, three things become available **at once**, without sequential effort:

NUMBER	WEIGHT	CONGRUENCE
How many relationships exist,	The strength or importance of	How relationships align, cluster, and

read by edge density
and node count at a
glance.

*"This region is busy;
that one is sparse."*

relationships, read by
edge thickness,
colour, label, shape.

*"These connections
are heavy; those are
incidental."*

point, read by spatial
proximity, group
shells, and shared
direction.

*"These belong
together; those two
clusters point at each
other."*

One does not *recall* the structure — one *sees* it. Reasoning is not sequential reconstruction; it is pattern recognition over a structure presented as a whole. That is a fundamentally higher-bandwidth way to grasp relational data than any linear medium can be, and it is the reason this tool exists.

Spatial memorability (Method-of-Loci-style "I remember the red book because it's over there") is a *secondary* effect that follows from the placement being stable across sessions. Useful, but not the primary thing.

2

Hand-authoring as sense-making

The act of placing a node *is* an act of reasoning. The author decides where it goes — and that decision is informed by their understanding of how it relates to the others. As more nodes are placed, the understanding sharpens; as the understanding sharpens, nodes move. The file is the residue of that thinking.

This is why hand-placement is non-negotiable (Axiom 1): if a layout algorithm overwrites the positions, it overwrites the reasoning. The author is not just *describing* a graph; they

are *thinking about it in 3D space*. A blank `.cns` file is an invitation to think. A finished one is the shape of a thought.

3

A persistent thinking artifact

A visual space lives across sessions. One can leave it and come back. One can evolve it as one's understanding evolves. One can revisit a layout from last year and ask "what was I thinking when I put that there?" — and often the placement itself encodes the answer.

"Bookmarks" (Axiom 4) operate on the short time-scale — saved viewpoints into a space for a particular act of perception. The space itself is the longer time-scale: a thinking-vehicle that outlasts any single sitting. A space is not a chart that captures a moment; it is a workshop the author keeps returning to.

03

AXIOMS

Treat these as design constraints on future work.

AXIOM 1

Placement is argument.

AXIOM 2

Hierarchical grouping

When the author places node A near node B and far from C, that placement **is** the claim "A and B are related; C is not." The space is the author's understanding of the relationships made directly perceptible. Anything that overwrites placement (auto-layout, repositioning on resize) erases the author's argument. Hand-placement is non-negotiable.

carries meta-relationships.

A `cluster` says "these N things are one *thing* at the next scale up." The cluster's own position then encodes how that aggregate relates to other aggregates. A `.cns` file is a multi-scale relational argument: each level reveals a different layer of structure. Treat cluster-level visuals (shell, tube frame, label) as carrying real reasoning content, not decoration.

AXIOM 3

Smooth motion is load-bearing, not polish.

A snap, spin, or teleport breaks the perceptual channel mid-act. One stops perceiving relationships and starts re-orienting from scratch. Smooth flight allows one to **sweep** across many relationships in a single continuous perceptual act, picking up patterns the static view can't show. Camera transitions are part of the medium, not decoration on top of it.

AXIOM 4

Bookmarks are saved arguments.

A bookmark is a curated viewpoint that frames a specific relationship-pattern for a specific perceptual act. The viewpoint **is** the argument. Bookmarks deserve first-class treatment (naming, organisation, easy travel) because each one is a piece of authored sense-making.

AXIOM 5

Edges carry as much

AXIOM 6

The app is a perception

Edges carry as much reasoning as nodes.

Edge thickness, colour, label, dash, arrowhead — these are the *weight* and *congruence* channels. The reasoning content of the graph lives as much in the edges as in the nodes. Invest in edge visual quality with the same seriousness as node visual quality.

The app is a perception instrument, not a navigator.

Its job is to be transparent enough that one forgets one is using it. Anything that draws attention to the app itself (lag, UI clutter, animation glitches, modal interruptions) competes with the perceptual act one is trying to perform. Default toward calm, get-out-of-the-way design.

AXIOM 7**A node is a handle onto content.**

A node's first-class job is to stand for — and open — a real file (usually markdown, sometimes a PDF). Its visuals (shape, colour, size) are secondary to that binding. Decoupling a node from the content it points at removes its reason to exist: the space is an interface to a corpus, not a diagram floating free of it.

04**GUIDING QUESTIONS
FOR FUTURE WORK**

When proposing a change, ask:

- 01 Does this serve **simultaneous perception** of relationships, or does it force the user into a sequential read?

- 02 Does this preserve the **perceptual channel during motion**, or does it interrupt it?

- 03 Does this respect the **author's placement** as argument, or does it overwrite it?

- 04 Does this strengthen one of the three perceptual signals (**number, weight, congruence**), or does it add unrelated noise?

- 05 Does this make the viewer **more transparent**, or more demanding of the user's attention?

- 06 Does this keep the space **bound to its corpus** — the real files its nodes stand for — or does it drift toward a diagram detached from the content?

- 07 Does this stay useful **across sessions and over time**, or is it tied to a transient state?

If a change scores well on these questions, it probably belongs. If it scores poorly, it is probably the **wrong design**.

Source of truth: [northstar.md](#). Markdown is canonical; this HTML is its presentation. When the substance changes, edit the Markdown first, then regenerate this sibling from it.

Visual language: mid-century modern hard-edge minimalism (after McLaughlin, Benjamin, Feitelson). Three accent colours — charcoal, mustard, persimmon — chosen once and used consistently throughout.

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